Livija Jakaite and Vitaly Schetinin
School of Computing and Engineering
University of Bedfordshire
Luton, UK

May 20, 2026

May 20, 2026

Editor-in-Chief
Bayesian Analysis
International Society for Bayesian Analysis

Dear Editor,

We are pleased to submit the manuscript

"Catalan-Exponential Priors for Bayesian Decision Trees: WAIC Consistency, Posterior Concentration, and PAC-Bayes Guarantees"

for consideration for publication in *Bayesian Analysis*.

Summary of contributions. This paper develops the theoretical foundations of the JBDT model (Jakaite & Schetinin, 2025, MDPI Make), a Bayesian decision tree classifier that uses a Catalan-exponential tree-size prior and Dirichlet-Multinomial leaf models with RJMCMC inference. We prove four main results that collectively provide a complete theoretical justification for the model design and explain previously unexplained empirical findings:

1. **WAIC-LOO gap theorem**: The WAIC and LOO deviances agree up to the constant $-4N/(N + \alpha_0) + O(N^{-1})$, proved in closed form. This yields the practical design criterion $N_{\min} \approx 5.41/\Delta$, which predicts the exact sample size at which WAIC-weighted BMA becomes reliable.
2. **Catalan prior tail bound**: $P(k \geq k_0) \leq C_\gamma \, (e^{-\gamma}/4)^{k_0}$, with universal effective decay rate $r_{\text{eff}} = 1.228 \, e^{-\gamma}/4$. Posterior concentration on the true complexity k^* is achieved at $N = 300$ (Catalan) versus $N > 800$ (Chipman 1998).
3. **BMA oracle inequality**: $R_{\text{BMA}} - R_{k^*} \leq \Delta_{\text{Jensen}} = O(\exp(-N\Delta_{\text{WAIC}}/2))$. Excess risk decreases 56-fold from $N = 20$ to $N = 800$, following $O(1/N)$.
4. **PAC-Bayes sample complexity**: Under the oracle posterior, the Catalan prior requires $8.1\times$ fewer samples than Chipman for sparse true models ($k^* = 1$), directly explaining its advantage in small-N medical imaging.

Supplementary appendices provide RJMCMC detailed balance, ECE calibration bounds, and decision-theoretic analysis connecting the above to asymmetric-loss classification. All results are accompanied by numerical verification on both synthetic and real knee osteoarthritis data.

Fit for Bayesian Analysis. The paper is squarely within the scope of *Bayesian Analysis*: it develops methodology for Bayesian model selection (WAIC) and BMA, analyses a novel prior construction (Catalan-exponential) analytically, and provides PAC-Bayesian guarantees. The application to medical imaging provides concrete motivation but the theoretical results are model-agnostic within the DM-leaf class.

Suggested associate editors / reviewers. Given the combination of Bayesian nonparametric trees, model selection, and PAC-Bayes bounds, We would suggest reviewers with expertise in Bayesian model selection, decision trees, or PAC-Bayesian theory.

Declaration. The manuscript is original, has not been published, and is not under consideration elsewhere. There are no conflicts of interest. The paper is **not anonymised** in this version; please advise if double-blind anonymisation is required for the current submission round.

Code availability. All experiments are fully reproducible. Code and pre-computed results are publicly available at `https://github.com/vitsch/jbdt`.

Yours sincerely,

Livija Jakaite and Vitaly Schetinin
School of Computing and Engineering
University of Bedfordshire, UK
`Vitaly.Schetinin@beds.ac.uk`
(corresponding)